

# Hybrid Green's Function Learning With Axial Reduction

A two-dimensional elliptic solver built from axial Green operators and a learned directional source split

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## REDUCE

a multi-dimensional elliptic problem  
into axial Green subproblems

## COUPLE

learn the directional source split that  
makes axial reconstructions  
consistent

## RECOVER

a multi-dimensional solution from  
coupled axial Green operators

# Axial Reduction of Elliptic Operators

Replace a global source-to-solution map with structured one-dimensional Green reconstructions

$$\begin{aligned} -\nabla \cdot (a \nabla u) + \mathbf{b} \cdot \nabla u + cu &= f & \text{in } \Omega, \\ u &= 0 & \text{on } \partial\Omega, \end{aligned} \quad \mathbf{b} = (b_x, b_y).$$

$$\mathcal{S}_{a,\mathbf{b},c} : f \mapsto u, \quad u = \mathcal{S}_{a,\mathbf{b},c}[f].$$

Operator fixed; source varies.

## DIRECT MAP

one global source-to-solution representation

High-dimensional and geometry-sensitive



## REDUCED VIEW

directional 1-D operators + learned coupling

line-wise inverses + learned source split

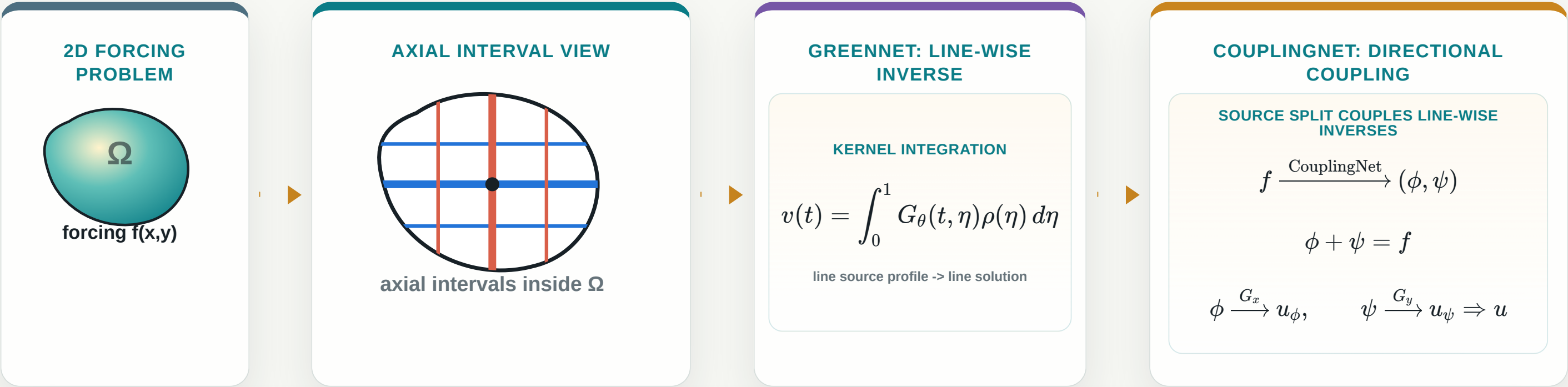
$$L_x u = -\partial_x(a \partial_x u) + b_x \partial_x u + \frac{1}{2}cu, \quad L_y u = -\partial_y(a \partial_y u) + b_y \partial_y u + \frac{1}{2}cu.$$

$$L_x u + L_y u = f.$$

GreenNet supplies line-wise Green inverses; CouplingNet learns the source split that turns them into a 2D elliptic solver.

# From a 2D Elliptic Problem to Coupled Axial Green Solves

Graphic abstract: slice, solve line-wise, learn the coupling, reconstruct the field



Axial GreenNet provides line-wise inverses; CouplingNet learns the directional source split that couples them into a 2D solution.

# Unit-Interval Pull-Back and Operator Scaling

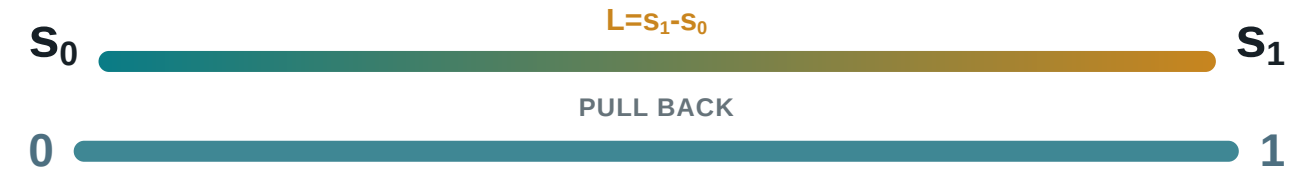
Physical length is absorbed into normalized operator coefficient profiles and source profile

## PHYSICAL ONE-DIMENSIONAL AXIAL OPERATOR

$$s \in [s_0, s_1],$$

$$\mathcal{L}_{\text{phys}} u = -\frac{d}{ds} \left( a_{\text{phys}}(s) \frac{du}{ds} \right) + b_{\parallel, \text{phys}}(s) \frac{du}{ds} + c_{\text{phys}}(s) u = f_{\text{phys}}(s).$$

## NORMALIZE THE COORDINATE



$$s = s_0 + Lt, \quad v(t) = u(s_0 + Lt), \quad t \in [0, 1].$$

The interval is normalized, but the physical length is not discarded.

$$a_{\text{unit}} = a_{\text{phys}}, \quad a'_{\text{unit}} = La'_{\text{phys}}, \quad b_{\parallel, \text{unit}} = Lb_{\parallel, \text{phys}}, \quad c_{\text{unit}} = L^2 c_{\text{phys}}, \quad f_{\text{unit}} = L^2 f_{\text{phys}}.$$

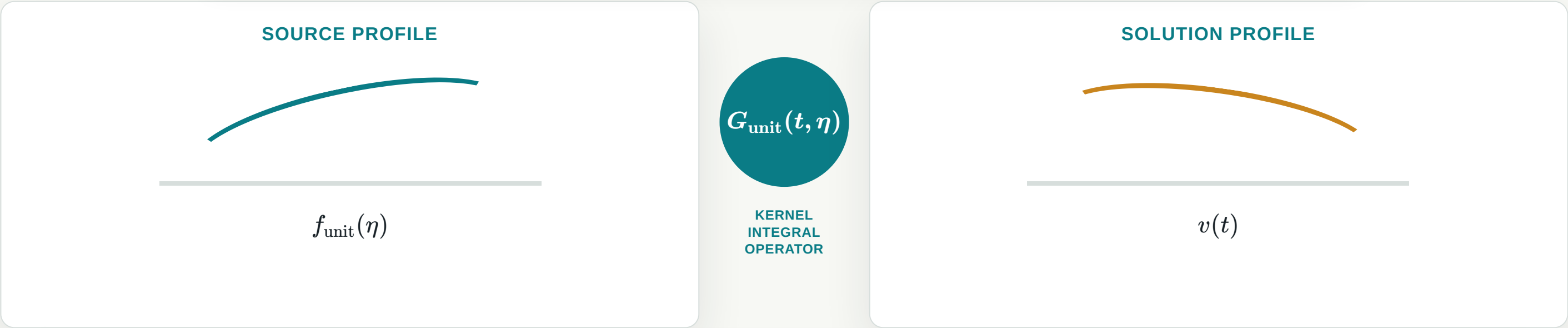
$$-\frac{d}{dt} \left( a_{\text{unit}}(t) \frac{dv}{dt} \right) + b_{\parallel, \text{unit}}(t) \frac{dv}{dt} + c_{\text{unit}}(t) v(t) = f_{\text{unit}}(t), \quad t \in [0, 1].$$

Here  $b_{\parallel} = b_x$  on an  $x$ -directed interval and  $b_{\parallel} = b_y$  on a  $y$ -directed interval.

# GreenNet I: Normalized Axial Green Operator

Learn one-dimensional Green kernels, not a global two-dimensional kernel

Previous step: normalize the operator. This step: learn its Green inverse.



$$\mathcal{G}_{\text{unit}}[f_{\text{unit}}](t) = \int_0^1 G_{\text{unit}}(t, \eta) f_{\text{unit}}(\eta) d\eta, \quad v = \mathcal{G}_{\text{unit}}[f_{\text{unit}}].$$

**Operator coefficients:** Axial coefficient profiles define each local 1D operator.

**Kernel coordinates**  $(t, \eta)$  define the evaluation and source points.

**GreenNet target** is a local unit-interval kernel, not a global 2D Green function.

# GreenNet II: Analytic Green Structure and Learned Correction

A hybrid kernel: Dirac delta structure, Heaviside cancellation, and learned smooth correction

Do not learn the Green singularity from scratch.

HYBRID GREEN KERNEL

$$G_{\theta} = \text{Dirac delta structure} + \text{Heaviside cancellation} + \text{learned smooth correction}.$$

**DIRAC DELTA STRUCTURE**

Build the source jump.

**HEAVISIDE CANCELLATION**

Cancel the induced Heaviside contribution.

**LEARNED SMOOTH CORRECTION**

Learn the smooth residual.

Analytic structure handles the singular terms before learning.

# GreenNet II: Analytic Green Structure and Learned Correction

A hybrid kernel: Dirac delta structure, Heaviside cancellation, and learned smooth correction

$$G_{\theta}(t, \eta) = \underbrace{E(t, \eta)M(t)R_{\theta}(t, \eta)}_{\text{learned smooth correction}} + \underbrace{B(t) \left( J_0(t, \eta) - \frac{1}{2}E(t, \eta) \right)}_{\text{Heaviside cancellation}} + \underbrace{A(t)G_0(t, \eta)}_{\text{Dirac delta jump}}.$$

## DIRAC DELTA STRUCTURE

$$G_0(t, \eta) = \begin{cases} t(1 - \eta), & t < \eta, \\ \eta(1 - t), & t \geq \eta, \end{cases} \quad \partial_t^2 G_0 = -\delta(t - \eta).$$

## HEAVISIDE CANCELLATION

$$\partial_t J_0(t, \eta) = G_0(t, \eta), \quad S = J_0 - \frac{1}{2}E, \quad \partial_t^2 S = \partial_t G_0.$$

Analytic structure handles the singular terms before learning.



# GreenNet II: Analytic Green Structure and Learned Correction

A hybrid kernel: Dirac delta structure, Heaviside cancellation, and learned smooth correction

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## LEARNED SMOOTH CORRECTION

$$E(t, \eta)M(t)R_{\theta}(t, \eta), \quad E(t, \eta) = t\eta(1 - \eta), \quad M(t) = 1 - t.$$

GreenNet learns  $R_{\theta}$ ;  $E(t, \eta)M(t)$  keeps that correction boundary-compatible. The singular and cancellation terms are already built into the kernel form.

The neural network learns the smooth residual with boundary-compatible envelope factors.



# GreenNet III: Source-to-Solution Supervision

Supervise the Green operator action, not pointwise exact-kernel labels

No pointwise Green-kernel labels.

$$w(t) \sim \mathcal{GP}(0, k_\ell)$$

→

$$v(t) = w(t) - ((1-t)w(0) + tw(1))$$

so  $v(0) = v(1) = 0$

→

$$f_{\text{unit}} = \mathcal{L}_{\text{unit}} v$$

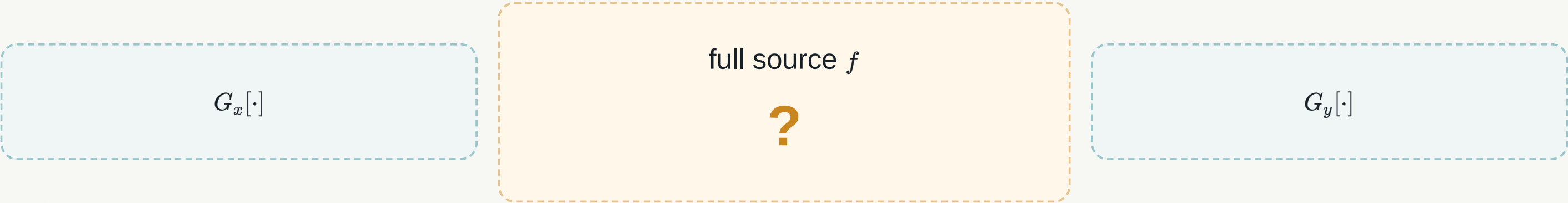
$$v_\theta(t) = \int_0^1 G_\theta(t, \eta) f_{\text{unit}}(\eta) d\eta.$$

$$\mathcal{J}_{\text{Green}} \sim \mathbb{E} \left[ \int_0^1 |v_\theta(t) - v(t)|^2 dt \right].$$

Training checks whether the learned Green operator maps sources generated from target solutions back to those target solutions.

# CouplingNet I: Directional Source Split

Learn how the forcing is divided between axial Green reconstructions



$$f \longmapsto (\phi, \psi).$$

**X-DIRECTED SOURCE**

$$\phi = L_x u, \quad L_x u = -\partial_x(a\partial_x u) + b_x \partial_x u + \tfrac{1}{2}cu.$$

**Y-DIRECTED SOURCE**

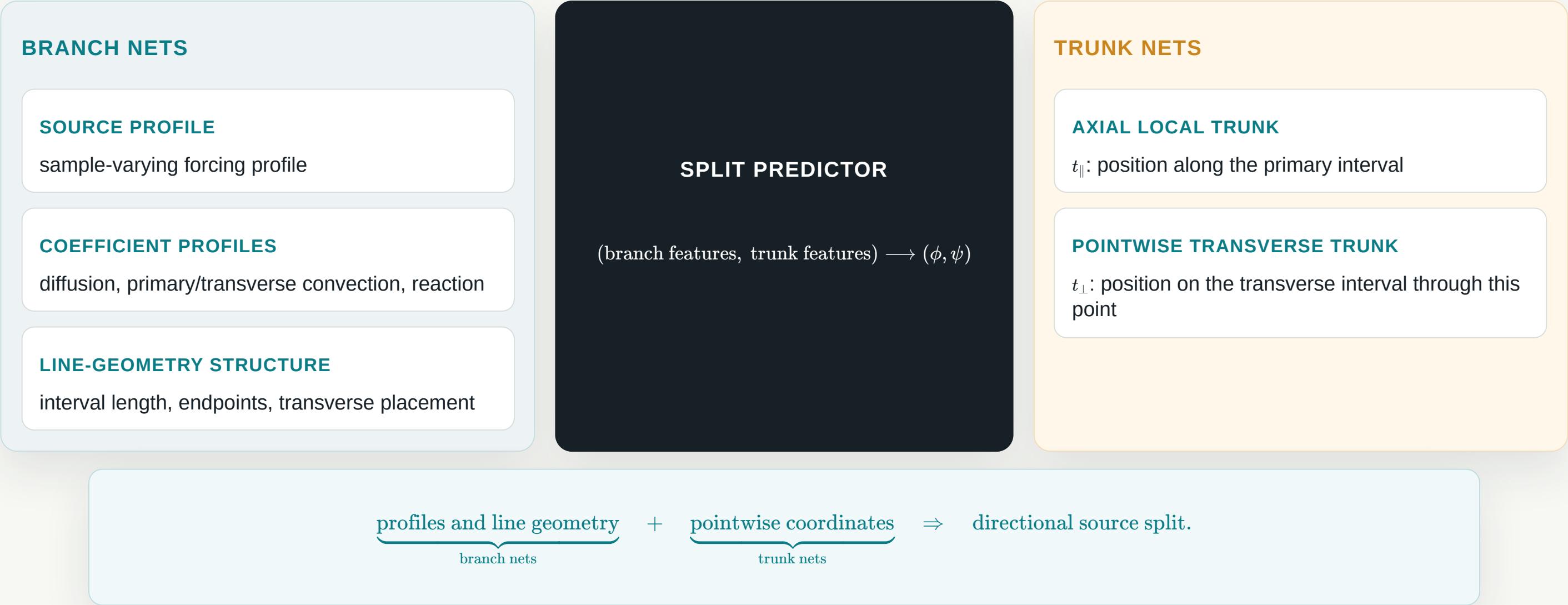
$$\psi = L_y u, \quad L_y u = -\partial_y(a\partial_y u) + b_y \partial_y u + \tfrac{1}{2}cu.$$

$$\phi + \psi = f.$$

CouplingNet learns line-wise flux-divergence/source components that couple the axial Green reconstructions.

# CouplingNet II: Branches and Local Context for Split Prediction

Branch nets encode profiles; trunk nets encode pointwise coordinates



# CouplingNet III: Projection and Green Reconstruction

Project in physical split variables, then reconstruct and average

## STAGE 1: PHYSICAL BALANCE PROJECTION

### COUPLINGNET RAW SPLIT

$$(\phi_{\text{raw}}, \psi_{\text{raw}})$$

not guaranteed balanced



### PROJECTION STEP

$$\begin{aligned} r &= f - (\phi_{\text{raw}} + \psi_{\text{raw}}) \\ \phi &= \phi_{\text{raw}} + \frac{1}{2}r, \quad \psi = \psi_{\text{raw}} + \frac{1}{2}r \end{aligned}$$



### BALANCED SPLIT

$$\phi + \psi = f$$

## STAGE 2: GREEN RECONSTRUCTION

### X-REPRESENTED SOLUTION

$$u_{\phi} = G_x[\phi]$$

### Y-REPRESENTED SOLUTION

$$u_{\psi} = G_y[\psi]$$

### FINAL AVERAGE

$$u_{\text{pred}} = \frac{1}{2}(u_{\phi} + u_{\psi})$$

$$(\phi_{\text{raw}}, \psi_{\text{raw}}) \xrightarrow{\text{projection}} (\phi, \psi), \phi + \psi = f \xrightarrow{G_x, G_y} (u_{\phi}, u_{\psi}) \xrightarrow{\text{average}} u_{\text{pred}}.$$

# Energy-Norm Error Bound Proposition

Small split energy controls the final energy error

$$u_\phi$$

$$u_{\text{pred}} = \frac{1}{2}(u_\phi + u_\psi)$$

$$u_\psi$$

DIFFUSION-WEIGHTED ENERGY NORM

$$\|v\|_a^2 := \int_{\Omega} a(x) |\nabla v(x)|^2 dx, \quad a(x) > 0.$$

Here  $a(x)$  is the diffusion coefficient.

SPLIT-ENERGY LOSS

$$\mathcal{E}_{\text{split}} = \|u_\phi - u_\psi\|_a^2.$$

REFERENCE SOLUTION

$$\mathcal{L}u_* = f, \quad u_*|_{\partial\Omega} = 0.$$

$$\|u_{\text{pred}} - u_*\|_a \leq \frac{C_E}{2} \sqrt{\mathcal{E}_{\text{split}}}.$$

$C_E$ : stability constant of the fixed elliptic operator.

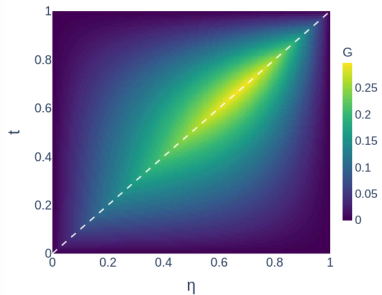
# Numerical Evidence I: GreenNet Kernel Approximation

Capturing the singular Green structure on an axial interval

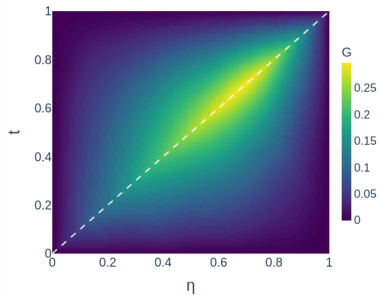
## KERNEL-LEVEL EVIDENCE

$$\Omega = \{x^2 + y^2 < 0.5^2\}, \quad -\nabla \cdot (a \nabla u) = f, \quad a(x, y) = 1 + \frac{1}{2} \sin(2\pi x) \sin(4\pi y), \quad \mathbf{b} = 0, \, c = 0.$$

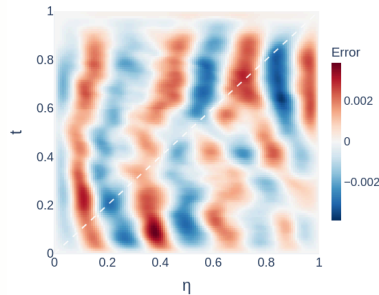
REFERENCE KERNEL



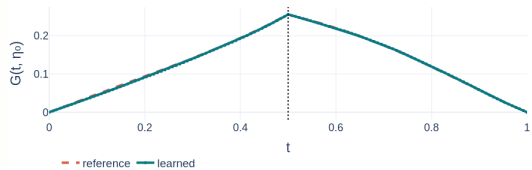
LEARNED KERNEL



SIGNED ERROR



FIXED-η SLICE



DIAGNOSTICS

|                            |                       |
|----------------------------|-----------------------|
| η                          | 0.50                  |
| Kernel rel. L2             | 1.11%                 |
| Slice rel. L2              | 0.92%                 |
| Diagonal band              | t-η  ≤ 5/128 = 0.0391 |
| Error mass / band area     | 9.5% / 8.3%           |
| Mean error / off-band mean | 1.16x                 |

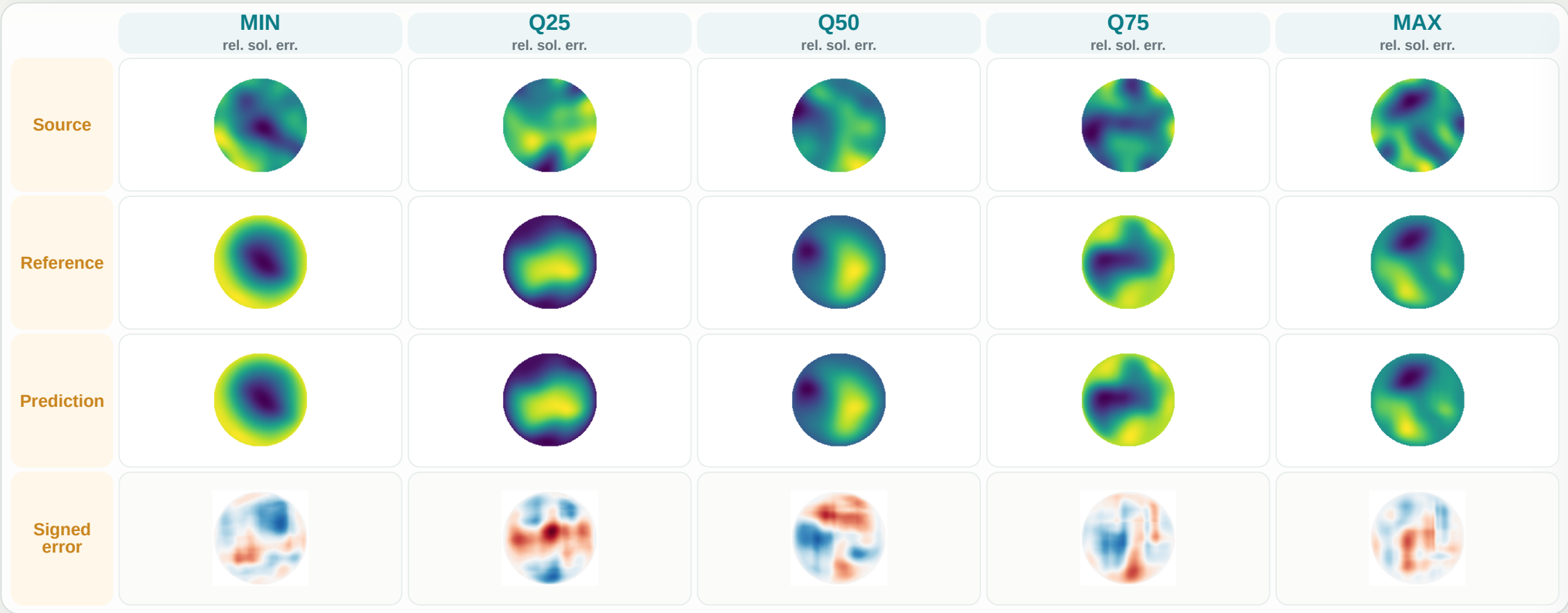
The learned kernel captures the singular Green structure, and the signed error is not concentrated near the singular diagonal.

# Numerical Evidence II: CouplingNet Solution Reconstruction

Quantile-selected samples show source, reference, prediction, and signed error

SOLVER-LEVEL EVIDENCE

$$\Omega = \{x^2 + y^2 < 0.5^2\}, \quad -\nabla \cdot (a \nabla u) = f, \quad a(x, y) = 1 + \frac{1}{2} \sin(2\pi x) \sin(4\pi y), \quad \mathbf{b} = 0, \, c = 0.$$



SELECTED BY RELATIVE SOLUTION ERROR

| CASE | REL. SOL. ERR. | SPLIT ENERGY |
|------|----------------|--------------|
| min  | 1.87%          | 7.48e-4      |
| q25  | 3.10%          | 8.59e-4      |
| q50  | 3.95%          | 7.41e-4      |
| q75  | 5.33%          | 6.87e-4      |
| max  | 8.84%          | 4.77e-4      |

Scales: per-sample source/solution; shared signed-error scale.

Quantile-selected samples show that the learned directional source split supports 2D solution reconstruction across the observed relative-error range, not only on a single favorable case.



# Takeaway: Coupled Axial Green Solvers

GreenNet learns line-wise Green kernels; CouplingNet learns the directional source split.

A 2D elliptic problem is solved through axial Green inversions and a learned, balance-preserving source decomposition.

## AXIAL GREEN KERNELS

Normalize axial intervals and learn line-wise inverse kernels.

## ANALYTIC STRUCTURE

Encode singular Green behavior before neural correction.

## SOURCE SPLIT

Learn  $\phi/\psi$  without reference-solution or split labels.

## ENERGY BOUND

Connect unsupervised split consistency to final solution error under assumptions.

GreenNet supplies line-wise Green inverses;  
CouplingNet learns the source split that turns them into a 2D elliptic solver.



# Backup / Q&A Menu

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Details available if the audience asks

## READY WITHOUT FIGURES

- Backup A: Dirac/Heaviside derivation sketch
- Backup B: imperfect Green reconstruction perturbation
- Backup C: connected-interval pull-back detail

## DEFERRED UNTIL FIGURES ARE SELECTED

- extra GreenNet reconstruction evidence
- extra CouplingNet split/energy evidence

# Backup A: Dirac/Heaviside Derivation Sketch

Why the analytic Green wrapping has three terms

Operator application creates two effects.

$$G_\theta = EMR_\theta + B \left( J_0 - \frac{1}{2} E \right) + AG_0.$$

### EFFECT 1: DIRAC JUMP

$$\partial_t^2 G_0(t, \eta) = -\delta(t - \eta)$$

The base term (A(t)G\_0(t,)) provides the required source singularity.

### EFFECT 2: HEAVISIDE LEFTOVER

$$\partial_t J_0(t, \eta) = G_0(t, \eta)$$

Applying the operator to the jump structure also creates a Heaviside-type contribution.

### ANALYTIC COMPENSATION

$$S = J_0 - \frac{1}{2} E, \quad \partial_t^2 S = \partial_t G_0$$

(B(t)S(t,)) cancels the Heaviside contribution before the neural correction is used.

$$A(t) = \frac{1}{a_{\text{unit}}(t)}, \quad B(t) = \frac{a'_{\text{unit}}(t) + b_{\parallel, \text{unit}}(t)}{a_{\text{unit}}(t)^2}.$$

The neural term (E(t,)M(t)R\_(t,)) learns the remaining smooth residual.

# Backup B: Imperfect Green Reconstruction Perturbation

What changes when the learned Green inverse is not exact?

$$G_x[\phi_*] = u_* + \varepsilon_x, \quad G_y[\psi_*] = u_* + \varepsilon_y.$$

$$\|u_{\text{pred}} - u_*\|_a \leq \frac{C_E}{2} \left( \sqrt{\mathcal{E}_{\text{split}}} + \|\varepsilon_x - \varepsilon_y\|_a \right) + \frac{1}{2} \|\varepsilon_x + \varepsilon_y\|_a.$$

## DIRECTIONAL MISMATCH

$$\varepsilon_x - \varepsilon_y$$

Visible to the split-consistency argument.

## COMMON BIAS

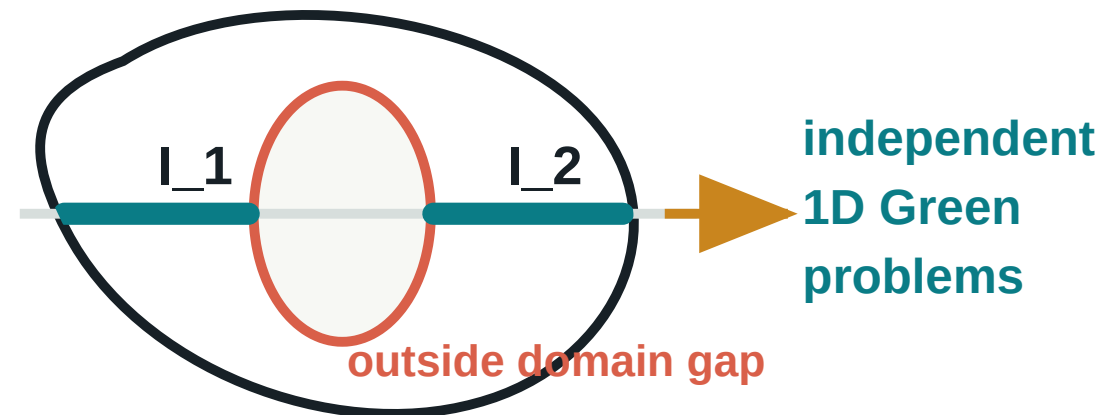
$$\varepsilon_x + \varepsilon_y$$

Can remain invisible when the two reconstructions agree.

# Backup C: Connected-Interval Pull-Back Detail

How non-square domains become normalized one-dimensional Green problems

ONE AXIAL SLICE CAN PRODUCE MULTIPLE CONNECTED INTERVALS



PULL BACK EACH CONNECTED INTERVAL SEPARATELY

$$I = [s_0, s_1], \quad s = s_0 + Lt, \quad t \in [0, 1].$$

$$a_{\text{unit}} = a_{\text{phys}}, \quad a'_{\text{unit}} = La'_{\text{phys}}, \quad b_{\parallel, \text{unit}} = Lb_{\parallel, \text{phys}}, \quad c_{\text{unit}} = L^2 c_{\text{phys}}, \quad f_{\text{unit}} = L^2 f_{\text{phys}}.$$

Each connected interval is an independent one-dimensional Dirichlet Green problem. Connected intervals are not merged across outside-domain gaps, because that would create artificial information flow outside the domain.